

Requirements Specification

Faculty Research Grant & Publication Tracker

PES University • Dept. of CSE • Lab 1: Requirements Engineering & UML Use-Case Modelling • Problem Statement #09 • SRN: PES1UG24AM193

1. Functional Requirements (FR-001 to FR-005)

ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall track grant fund allocations, allow faculty to log equipment procurement expenses, and compute remaining budgetary balance in real time.	High	Pass: Balance updates accurately and blocks expenses exceeding grant cap. Fail: Overdraft allowed without admin override.	Prevents accidental or unauthorised overspending of sponsored grant funds and keeps the Faculty Researcher aware of remaining budget at all times.
FR-002	The system shall allow faculty to register publication records with title, venue, indexing database (Scopus/SCI/WoS) and citation count, and shall refresh citation counts on a scheduled sync.	High	Pass: New publication appears on the dashboard with correct indexing tag within 24 hours of entry. Fail: Publication saved with a blank indexing or citation field.	Publication metrics are the primary KPI the Research Dean reports to funding and accreditation bodies each cycle.
FR-003	The system shall route every newly added co-author on a publication to that co-author for digital approval before the publication record is marked final.	Medium	Pass: Publication status becomes "Verified" only after all listed co-authors approve. Fail: Publication is finalised with a pending or rejected co-author.	Protects authorship integrity and prevents disputes that could surface during external research audits.
FR-004	The system shall generate a fund burn-up chart for each grant showing cumulative spend against the sanctioned budget, filterable by expense category and month.	High	Pass: Chart renders with cumulative totals matching the underlying ledger entries. Fail: Displayed total deviates from the ledger sum by any amount.	Lets the Research Dean monitor utilisation across multiple concurrent grants and flag under- or over-spending ahead of renewal deadlines.
FR-005	The system shall automatically notify the Faculty Researcher and Research Dean when cumulative expenses on a grant cross 80% of the sanctioned amount.	Medium	Pass: An in-app/email alert is triggered exactly once when the 80% threshold is crossed. Fail: No alert is sent, or the alert repeats on every login.	Provides an early warning that complements the hard cap in FR-001, giving the researcher time to re-plan spending before funds are exhausted.

2. Non-Functional Requirements (NFR-001 & NFR-002)

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The system shall maintain an immutable, tamper-evident audit ledger for all financial approvals and journal status modifications.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Any logged approval can be altered or deleted without a traceable audit entry.	Sponsored-research funds and publication records are subject to statutory/institutional audit, so every change must be provably genuine.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-002	Availability & Usability	The system shall be available at least 99.5% of the time during business hours and shall render correctly on both desktop and mobile browsers (360px-1920px width).	Medium	Pass: Uptime monitoring confirms ≥99.5% monthly availability and layout remains usable across the tested screen widths. Fail: Monthly downtime exceeds 0.5% or the layout breaks on any tested device.	Faculty and the Dean often check dashboards close to grant renewal or publication-reporting deadlines, so reliable anytime access is essential.

Actors: Faculty Researcher, Research Dean | Domain: Campus & Academic Operations